neo4j网页http://localhost:7474/browser/

密码zhangyu123，用户名是默认的neo4j，其他不用填

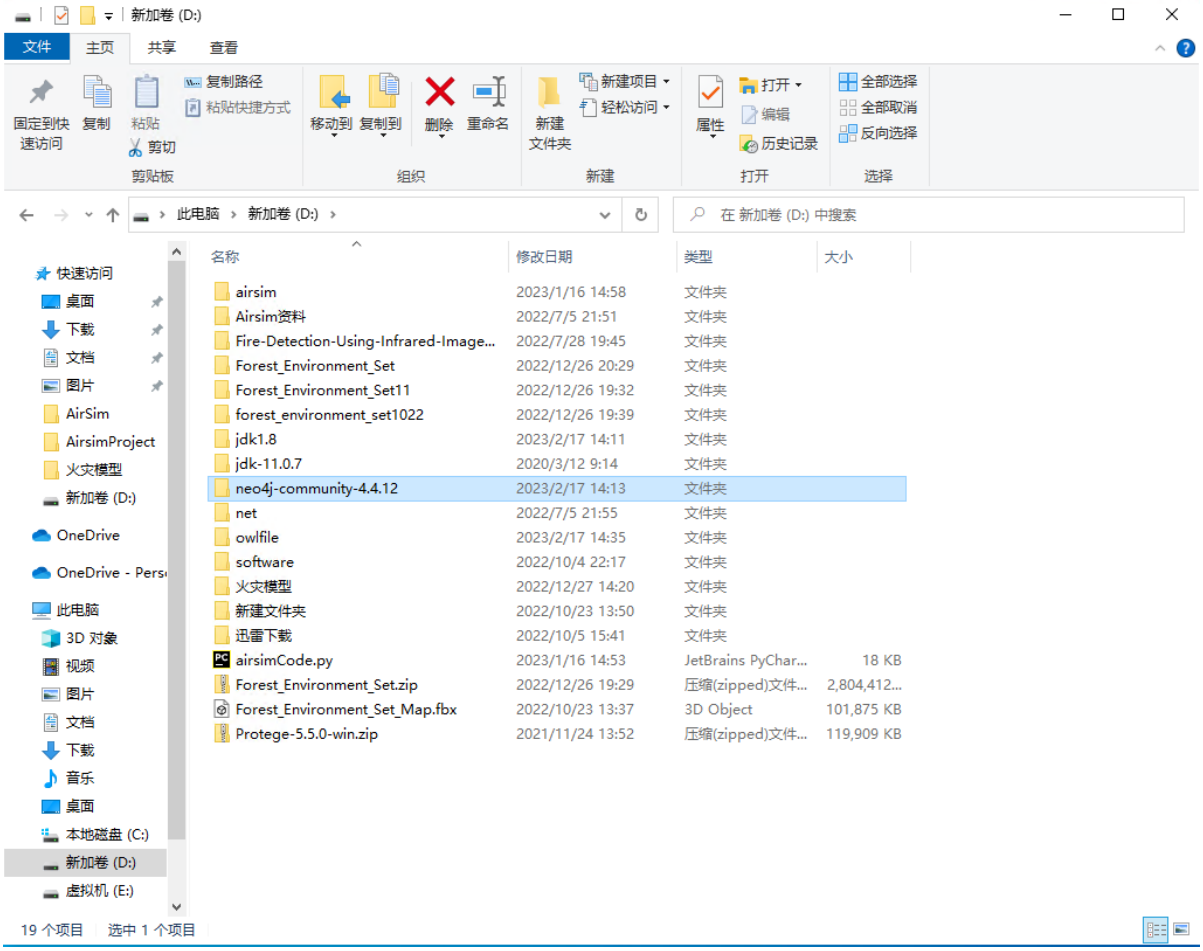
前期准备(已经做了)

1.下载扩展neosemantics jar包，将jar复制到neo4j/plugins目录下(已经做了)

https://github.com/jbarrasa/neosemantics/releases/download/3.3.0.2/neosemantics-3.3.0.2.jar

注意版本匹配

服务器上neo4j路径如下：



进去之后的目录下有个新建文本文档.txt，建议先看一下

2.修改配置文件(已经做了)

在neo4j/neo4j.conf文件中添加以下内容：

dbms.unmanaged_extension_classes=semantics.extension=/rdf

3.重新启动 neo4j，有这个就对了(已经做了)

```
Starting Neo4j.
2022-11-02 09:46:20.479+0000 INFO  Starting...
2022-11-02 09:46:21.114+0000 INFO  This instance is ServerId{4a4fdeaa} (4a4fdeaa-16cb-455f-a462-aed905b44f3d)
2022-11-02 09:46:22.048+0000 INFO  ======== Neo4j 4.4.12 ========
2022-11-02 09:46:28.853+0000 INFO  Performing postInitialization step for component 'security-users' with version 3 and
status CURRENT
2022-11-02 09:46:28.853+0000 INFO  Updating the initial password in component 'security-users'
2022-11-02 09:46:29.782+0000 INFO  Bolt enabled on 127.0.0.1:7687.
2022-11-02 09:46:29.794+0000 INFO  Mounted unmanaged extension [semantics.extension] at [/rdf]
2022-11-02 09:46:30.385+0000 INFO  Remote interface available at http://localhost:7474/
2022-11-02 09:46:30.388+0000 INFO  id: 986A00ACE6177385A3DC0AEF956F1A1D7AFE84FA4350FF2ABA7BB731FA7470D9
2022-11-02 09:46:30.388+0000 INFO  name: system
2022-11-02 09:46:30.388+0000 INFO  creationDate: 2022-11-02T09:20:14.854Z
2022-11-02 09:46:30.388+0000 INFO  Started.
```

切换neo4j默认数据库需要去配置文件修改配置

启动neo4j

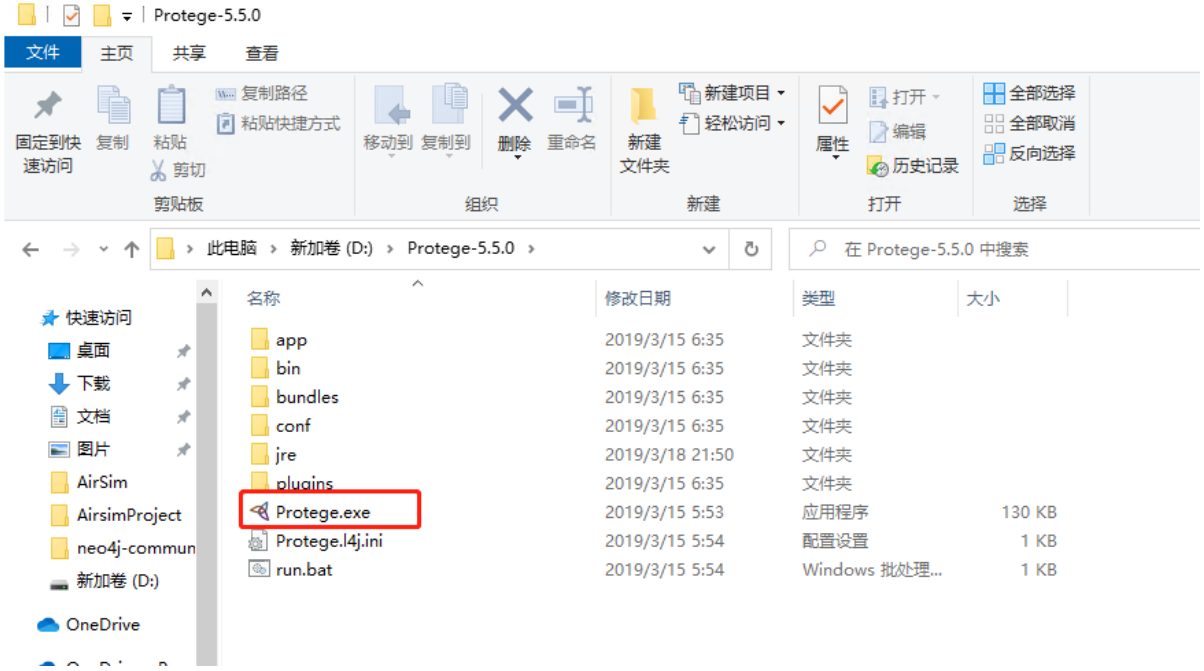
打开windows terminal

```
进入D:\airsim目录下，找到neo4j-community-4.4.12文件夹，进去
d:
cd .\neo4j-community-4.4.12\
cd .\bin\
neo4j console
```

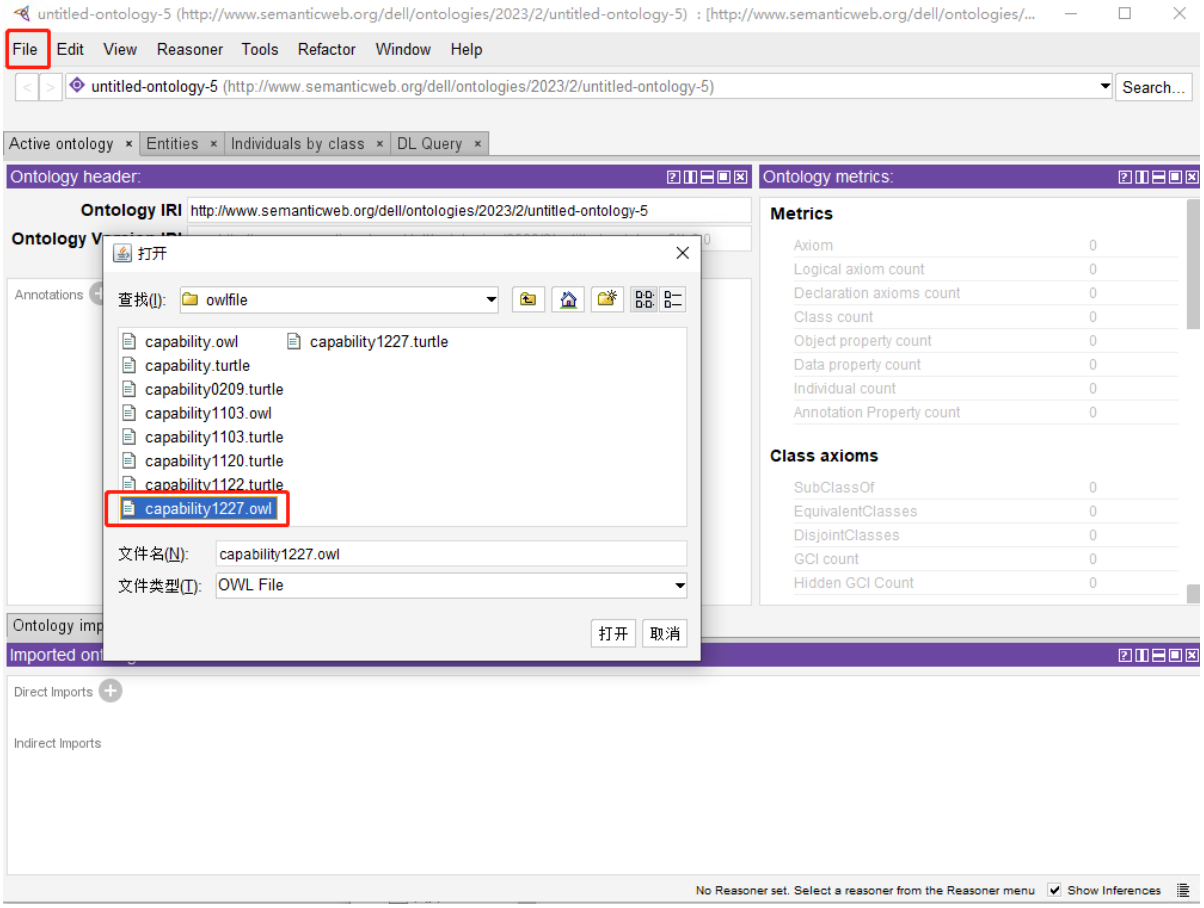
修改能力知识图

0.拉取仓库中的owlfile

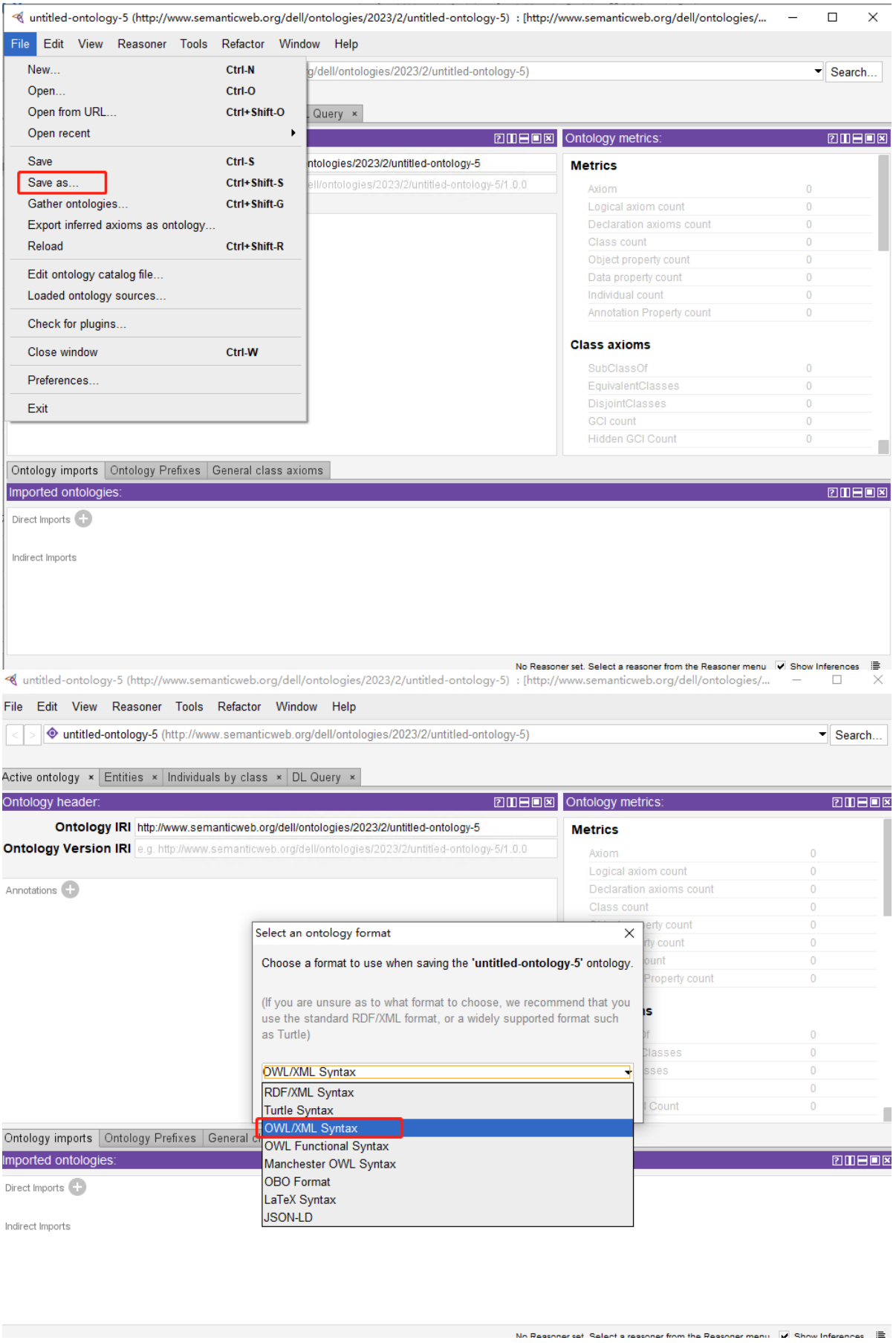
1.打开protege软件



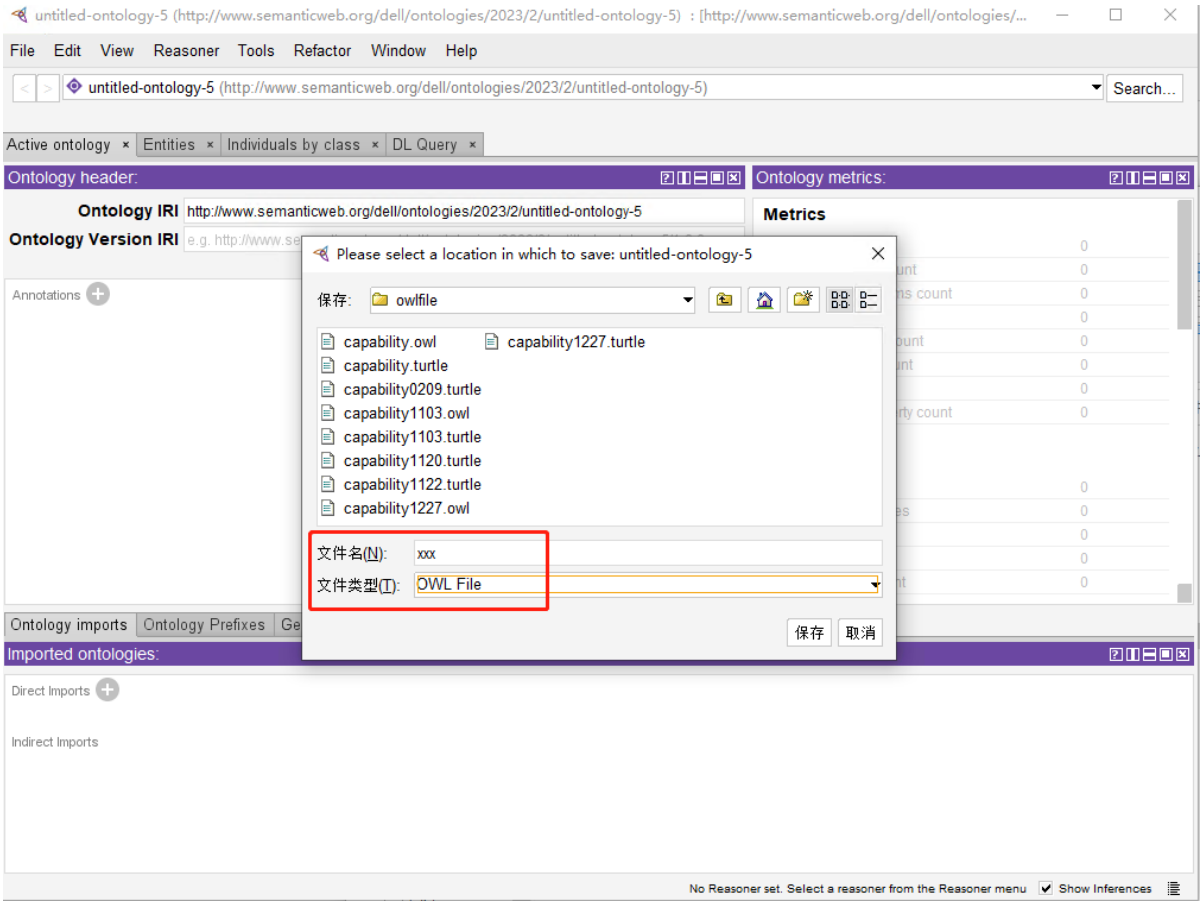
2.打开知识图的owl源文件



3.修改你要修改的任何东西，然后另存为新的.owl文件，如果你中途按了ctr + s保存了，那么也就没必要另存为了。可以参考教程https://blog.csdn.net/Jenny_oxaza/article/details/83148300学习protege。


untitled-ontology-5 (http://www.semanticweb.org/dell/ontologies/2023/2/untitled-ontology-5) : [http://www.semanticweb.org/dell/ontologies/...
File Edit View Reasoner Tools Refactor Window Help
New... Ctrl-N
Open... Ctrl-O
Open from URL... Ctrl+Shift-O
Open recent
Save Ctrl-S
Save as... Ctrl+Shift-S
Gather ontologies... Ctrl+Shift-G
Export inferred axioms as ontology...
Reload Ctrl+Shift-R
Edit ontology catalog file...
Loaded ontology sources...
Check for plugins...
Close window Ctrl-W
Preferences...
Exit
Search...
Ontology metrics:
Metrics
Axiom 0
Logical axiom count 0
Declaration axioms count 0
Class count 0
Object property count 0
Data property count 0
Individual count 0
Annotation Property count 0
Class axioms
SubClassOf 0
EquivalentClasses 0
DisjointClasses 0
GCI count 0
Hidden GCI Count 0
Ontology imports
Ontology Prefixes
General class axioms
Imported ontologies:
Direct Imports
Indirect Imports
No Reasoner set. Select a reasoner from the Reasoner menu
Show Inferences

untitled-ontology-5 (http://www.semanticweb.org/dell/ontologies/2023/2/untitled-ontology-5) : [http://www.semanticweb.org/dell/ontologies/...
File Edit View Reasoner Tools Refactor Window Help
untitled-ontology-5 (http://www.semanticweb.org/dell/ontologies/2023/2/untitled-ontology-5)
Search...
Active ontology
Entities
Individuals by class
DL Query
Ontology header:
Ontology IRI http://www.semanticweb.org/dell/ontologies/2023/2/untitled-ontology-5
Ontology Version IRI e.g. http://www.semanticweb.org/dell/ontologies/2023/2/untitled-ontology-5/1.0.0
Annotations
Ontology metrics:
Metrics
Axiom 0
Logical axiom count 0
Declaration axioms count 0
Class count 0
Select an ontology format
Choose a format to use when saving the 'untitled-ontology-5' ontology.
(If you are unsure as to what format to choose, we recommend that you use the standard RDF/XML format, or a widely supported format such as Turtle)
OWL/XML Syntax
RDF/XML Syntax
Turtle Syntax
OWL/XML Syntax
OWL Functional Syntax
Manchester OWL Syntax
OBO Format
LaTeX Syntax
JSON-LD
Ontology imports
Ontology Prefixes
Imported ontologies:
Direct Imports
Indirect Imports



另存为owl之后就修改完了，下面需要导入neo4j

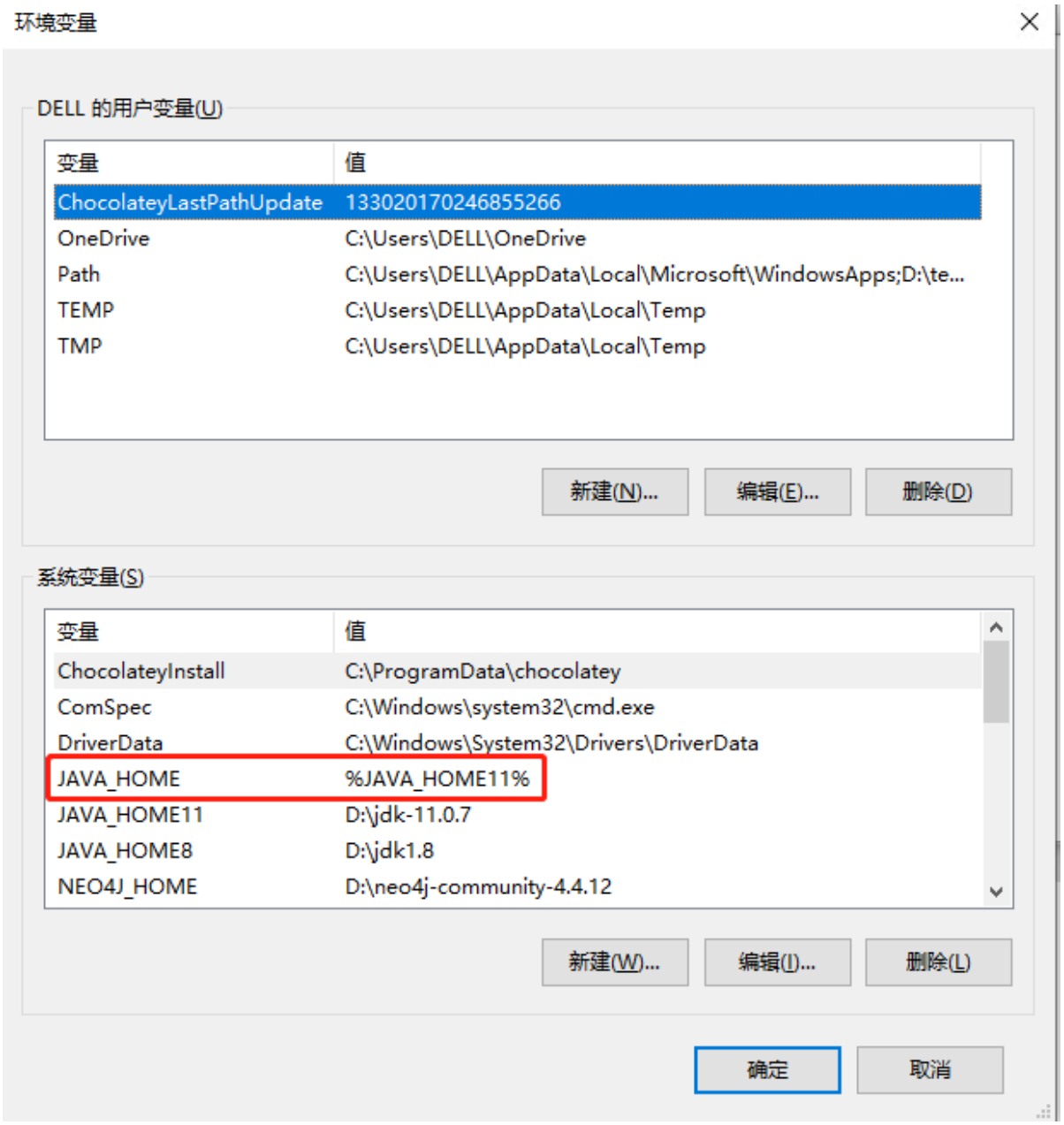
如果修改了owl文件，需要owl to turtle，以导入neo4j

1.切换jdk8

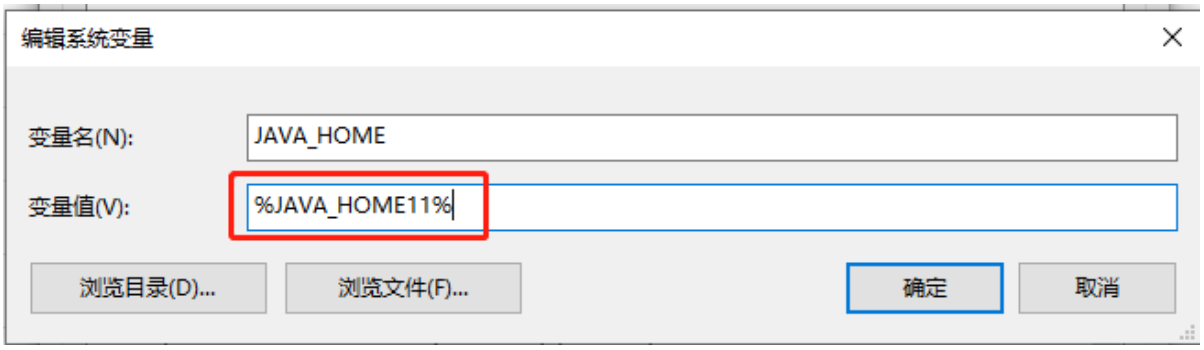
此电脑，属性



进去之后选择环境变量



修改红框为8即可



2.下载jar包（已经下了）

http://www.l3s.de/~minack/rdf2rdf/

3.转换为.turtle

owl文件跟 rdf2rdf-1.0.1-2.3.1.jar包放在同一个目录下，最好英文，然后执行命令

```
java -jar rdf2rdf-1.0.1-2.3.1.jar capability.owl capability.turtle
```

得到turtle文件，检查turtle文件大小是否正常。0k则为转换失败。

导入turtle到neo4j

1.切换jdk11，参考上面切换到8，只需要改为11即可。

2.（可选）新建或切换数据库，打开neo4j.conf配置文件，修改为你的数据库（一般来说不需要，除非你想备份之前的图数据库）

```
dbms.default_database = capability0206
```

3.启动neo4j

4.打开网站http://localhost:7474/browser/删除以往的图（没有做第2步的话，之前的数据库就被覆盖了）

```
MATCH (n) DETACH DELETE n
```

5.配置环境，执行查询

```
CREATE CONSTRAINT n10s_unique_uri ON (r:Resource) ASSERT r.uri IS UNIQUE
call n10s.graphconfig.init()
```

6.执行查询，导入文件，注意修改文件路径和.turtle文件名

```
call n10s.rdf.import.fetch( "file:///E:/neo4j/owlfile/capability.turtle", "RDF/XML",
{handleVocabUris: "IGNORE"})
```

或者

```
CALL semantics.importRDF('file:///E:/neo4j/owlfile/capability.turtle', 'RDF/XML',{})
```

7.检查是否成功，执行命令

```
MATCH (n) return n
```

去掉无用的uri（非必要）

我的uri是http://www.semanticweb.org/zhengyanzhu/ontologies/2022/8/untitled-ontology-3#，长度为77（字符个数），建议在做owl的时候改短一点

```
match (n) where n.uri=~"http://www.semanticweb.org/zhengyanzhu/ontologies/2022/8/untitled-ontology-3#.*" set n.uri=substring(n.uri,77) return n
```